

Jaipreet Singh Sandhu

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EDUCATION

Chandigarh University | Mohali, Punjab

Bachelor of Engineering in Computer Science | GPA: 6.98

August 2028

TECHNICAL PROJECTS

AI-Powered EdTech Platform with RAG Chatbot | Personal Project

- Built a full-stack EdTech Platform using Spring Boot, Angular, and PostgreSQL with JWT authentication and role-based access control for secure management of students, courses, and educational content.
- Built a RAG chatbot using LangChain4j, Ollama (DeepSeek-R1), and PostgreSQL pgvector, enabling students to query teacher-uploaded course materials with answers grounded strictly in the provided content.
- Integrated AWS S3 for content storage and built a Dockerized ingestion pipeline with PDF parsing, text chunking, and embedding generation for scalable deployment.

Face Recognition Attendance Management System | Personal Project

- Utilized the machine learning-based face_recognition library (built on dlib) for face encoding and matching, alongside OpenCV for camera integration and image processing, with Tkinter GUI and SQLite3 database.
- Added student ID search, real-time attendance marking, and full data reset functionality for testing cycles.

SKILLS

Languages - C++, Python, Java, SQL

Frameworks/Technologies - Spring Boot, FastAPI, Angular, React, Docker, LangChain4j, MongoDB, PostgreSQL

Concepts - OOP, Data Structures, DBMS, Problem Solving

PROFESSIONAL EXPERIENCE

National Informatics Centre (NIC) – National Digital Recruitment Platform (NDRP) | Client: National Recruitment Agency (NRA)

Backend Developer Trainee | Dec 2025 – Jan 2026

- Developed RESTful APIs using FastAPI (Python)
- Implemented backend features and optimized system performance
- Debugged production issues and improved deployment understanding

CERTIFICATIONS

- Python for Data Science, AI & Development | IBM, Coursera
- Python Data Analytics | Meta, Coursera
- Introduction to Software Engineering | IBM, Coursera